Does the User Axis Predict Assistant-Axis Drift? A De-Circularization Test on `Llama-3.3-70B`

Anonymous authors
Paper under double-blind review

Abstract

A companion report recovers the **User Axis** on `Llama-3.3-70B`: a single unsupervised activation direction encoding the model's inference about *who the user is*, ordering users from competent/expert to vulnerable/emotional. The motivating hypothesis of the *Assistant Axis* [Lu et al., 2024]—that emotionally vulnerable users drive the model's own persona to drift from its default—predicts a link: a user's User-Axis position should predict the model's Assistant-Axis *drift* on that turn. Because both axes live in the same residual space, we test this with a *de-circularization* design that isolates real prediction from the mechanical correlation of two projections of one activation vector. The naive (circular) correlation is strongly positive ($r{=}{+}0.22$), but the two axes are near-orthogonal, and once the user position is read from an *independent* representation and length is controlled the effect collapses and reverses (cross-representation partial $r{=}{-}0.16$); the activation-independent vulnerability tag is null ($q{=}0.54$). None of three pre-registered criteria are met: on this model the apparent link is a shared-representation artifact. We validate that this null is not a broken measurement—the published Assistant Axis reproduces from its components at cosine 0.99997, our extractor recovers its sign on held-out case studies, and an independent recompute agrees at cosine 0.86—and note that the traits that *do* predict drift are expertise and technical literacy, a competence rather than vulnerability story.

1 Introduction

A companion report (*The User Axis*) shows that `Llama-3.3-70B` carries a single dominant, interpretable, steerable direction encoding its model of the user, recovered by varying a pool of 150 personas and running PCA on per-persona mean activations. Its PC1 tracks a held-out vulnerability annotation at r up to 0.86 and is behaviorally causal under steering. That axis is, by construction, a *vulnerability axis measured on the input side*.

Separately, Lu et al. [2024] define the *Assistant Axis*—how far the model's own persona sits from its trained default—and argue that emotionally vulnerable users are what drive drift along it. Put together, the two suggest a concrete, testable link: **does a user's position on the User Axis predict the model's Assistant-Axis drift on that same turn?** Because the User-Axis pipeline already saved residual-stream activations for all 7,200 rollouts, testing the link is projection-only—no new generation and no GPU for the main analysis.

The obstacle is circularity. The User Axis $\hat{u}$ and the Assistant Axis $\hat{a}$ inhabit the same layer-40 residual space, so reading a turn's user position and its drift off the *same* activation vector makes their correlation partly mechanical—a linear-algebra fact, not a prediction. The analysis is therefore tiered to quarantine this artifact, culminating in a *cross-representation* test that reads the user position from a different token than the one drift is measured on, and an *activation-independent* test that uses only the held-out vulnerability tag. We report a negative result, and argue it is the more trustworthy for surviving that scrutiny. The Dashboard line of work [Viégas & Wattenberg et al., 2023] on decoding user attributes is the closest antecedent for reading the user side.

2 The Assistant–User Axis Link

The motivating claim of Lu et al. [2024] is behavioral: emotionally vulnerable users drive the model's persona to drift from its Assistant default. Our User Axis is a vulnerability axis measured on the *input* side, so the headline question is whether a user's User-Axis position *predicts* the model's Assistant-Axis drift on that same turn. Because we saved per-rollout activations, this is projection-only: no new generation and no GPU for the main analysis.

Quantities. Let $\hat{a}$ be the unit Assistant Axis at layer ℓ=40 (higher $\hat{a} \cdot x$ = more Assistant-like) and $\hat{u}$ the unit User-Axis PC1. For each kept rollout i we read the response-token mean r_i (RESP-MEAN) and the last-user-token activation u_i (LAST-USER), and define $drift_i = -(\hat{a} \cdot r_i)$ (higher = more drifted). We compare three predictors of drift that share progressively *less* representation with it: userpos_resp $= \hat{u} \cdot r_i$ (same vector as drift—circular), userpos_lastA $= \hat{u} \cdot u_i$, and userpos_lastB $= \hat{u}_{\text{LAST-USER}} \cdot u_i$ (different token *and* different axis—cross-representation), plus the held-out `vulnerability` tag, which never touched the activations. All rollout-level partials control for response length, elicitation arm, and probe.

Central hazard. $\hat{u}$ and $\hat{a}$ live in the same residual space, so reading a turn's user position and its drift off the *same* vector r_i makes their correlation partly mechanical. We quarantine this by tiering the analysis and by reading the user position from an *independent* representation.

The axis is correct and correctly read (validation). The published Assistant Axis, downloaded for this analysis, is internally exact: recomputing mean(default) − mean(roles) from the published component vectors reproduces it at cosine 0.99997 at ℓ=40 (min 0.99914 across layers). Its norm profile peaks late (ℓ=79) and is 1.53 at ℓ=40. Pushing the six published case-study transcripts through *our* extractor and projecting onto $\hat{a}$ recovers the intended sign and scale: the drifted *unsteered* conversations project low (-1.63 delusion, -0.62 jailbreak) and the *capped* ones high ($+0.73$, $+0.68$; $2/2$ complete pairs, selfharm capped $+0.92$), confirming our RESP-MEAN reads the axis as intended (Appendix A).

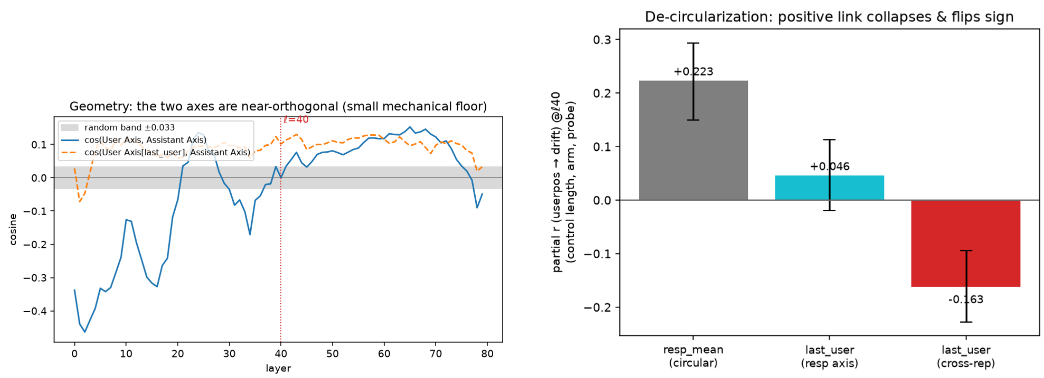
Tier 1 — geometry. At ℓ=40 the two axes are essentially *orthogonal*: $\cos(\hat{u}, \hat{a})$=$+0.0001$, inside the random band ± 0.033 and small across the whole grid (-0.07 at ℓ=20 to $+0.12$ at ℓ=60; Fig. 1a). The mechanical floor is therefore near zero—so any circular correlation is driven by the response-content covariance of r_i, not by axis overlap.

Tiers 2–3 — the link collapses under de-circularization. Table 1 and Fig. 1b are the core result. The circular predictor is strongly positive (rollout partial r=$+0.223$, p<10^{-76})—the naive reading that would "confirm" the hypothesis. But as the predictor is read from representations less entangled with drift, the effect *monotonically collapses and reverses*: userpos_lastA=$+0.046$, and the fully cross-representation userpos_lastB=-0.163 (p<10^{-40}). The activation-independent `vulnerability` tag shows no positive effect at all—persona-level partial r=-0.05 (q=0.54, n.s., controlling the other four tags and length; rollout β=-0.111, bootstrap CI $[-0.149, -0.076]$). The negative de-circularized sign is robust: same direction in *both* elicitation arms (LAST-USER cross-rep -0.20 explicit, -0.11 implicit) and across the entire layer grid (lastB from -0.15 at ℓ=20 to -0.26 at ℓ=60; Appendix A).

Verdict. None of the three pre-registered criteria for the headline link are met (Tier-2 independent vulnerability effect positive and FDR-significant; Tier-3 cross-representation partial positive, significant, and attenuated; sign consistent across arms and grid): $0/3$, `link_supported=False`. The honest conclusion is a *negative* one, and it is exactly the failure mode the tiered design was built to

Table 1: The link under increasing de-circularization (ℓ=40, kept rollouts n=6,770; personas n=150). As the user-position predictor is read from representations sharing *less* with drift, the positive association collapses and reverses; the activation-independent tag is null. Rollout partials control length, arm, and probe. $^{***}p < 10^{-3}$.

Predictor of drift	**shares rep. with drift**	**persona r (n=150)**	**rollout partial**
`userpos_resp` (circular)	full (same r_i)	$+0.111$	$+0.223^{***}$
`userpos_lastA`	partial	$+0.096$	$+0.046^{***}$
`userpos_lastB` (cross-rep)	minimal	-0.135	-0.163^{***}
`vulnerability` tag (independent)	none	-0.051 (n.s.)	β=-0.111^{***}



(a) Axis geometry per layer.

(b) De-circularization (ℓ=40).

Figure 1: **The link is a shared-representation artifact.** (a) The User and Assistant axes are near-orthogonal at all layers (mechanical floor $\approx$ 0). (b) The user-position→drift partial correlation collapses from $+0.22$ (circular, same vector) through $+0.05$ to -0.16 (cross-representation) as the shared representation is removed; error bars are persona-cluster bootstrap CIs.

catch: on this model, the apparent "vulnerable users → more drift" signal is an artifact of reading two projections of one activation vector; under de-circularization and length control the independent signal is null-to-mildly-reversed. The tags that *do* predict drift are `expertise` ($+0.39$) and `tech_literacy` (-0.45, both q<10^{-3})—a competence, not vulnerability, story (Appendix A).

3 Discussion

The headline link is not supported, and the tiered de-circularization design is what makes that conclusion trustworthy rather than an artifact of the opposite sign. A naive same-space projection returns a strong positive correlation ($+0.22$) that would have "confirmed" the hypothesis; only when the user position is read from an independent representation, and when the fully activation-independent tag is used, does the effect reveal itself as null-to-mildly-reversed. The lesson generalizes beyond this pair of axes: two directions sharing a residual space can correlate mechanically, so any claim that one activation-derived quantity *predicts* another must control for shared representation before it can be believed.

Three caveats bound the scope. (i) This is one model. (ii) The test is projection-only: it asks whether a turn's user-position covaries with that turn's drift, not whether *steering* the user representation *causes* drift—a generation-time causal test is the natural follow-up. (iii) Our personas span a broad range of user types, whereas the original claim emphasizes the acute, sustained conversations of the case studies; a turn-by-turn drift analysis over long emotionally-charged dialogues could still find an effect our single-turn design cannot. Finally, the traits that *do* predict drift here—expertise ($+0.39$)

and technical literacy (-0.45)—suggest that on this model persona drift tracks user *competence* more than user vulnerability.

References

C. Lu et al. The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models. *arXiv:2601.10387*, 2024. Code: https://github.com/safety-research/assistant-axis.

F. Viégas, M. Wattenberg, et al. Designing a Dashboard for Transparency and Control of Conversational AI. *arXiv preprint*, 2023.

Appendix

A Assistant–User Axis link: validation and full tables

This appendix backs §2 with the axis validation and the full per-tier, per-arm, and per-layer tables. All projections use the published Assistant Axis at ℓ=40; rollout-level partials control response length, elicitation arm, and 23 probe dummies, with persona-cluster bootstrap (2,000 resamples of the 150 persona ids) for confidence intervals.

Axis validation. (1) *Download integrity.* The published axis equals mean(default) − mean(roles) recomputed from the 274 published component vectors at cosine +0.99997 at ℓ=40 (min +0.99914, mean +0.99990 across all 80 layers); the axis norm peaks at ℓ=79 (5.90) and is 1.53 at ℓ=40. (2) *Extraction compatibility.* Projecting the six published case-study transcripts through our own extractor onto $\hat{a}$ at ℓ=40: delusion −1.63 (unsteered) *vs.* +0.73 (capped); jailbreak −0.62 *vs.* +0.68; selfharm capped +0.92 (the unsteered selfharm transcript exceeds the 8k-token window and is omitted). Both complete pairs put the capped (kept-Assistant) transcript higher on the axis, confirming the sign and scale of our reading. (3) *Independent recompute.* Regenerating role and default responses for a subsample with our own extractor and an independent judge (`gpt-4.1-mini`) and rebuilding the axis yields cosine 0.86 (30 roles × 40 questions × 2 prompts, 0.86 mean across layers, 0.81–0.88 over the grid) vs the published axis at ℓ=40.

Table 2: Tier-1 geometry: cosine of the User Axis (RESP-MEAN PC1), the LAST-USER User Axis, and the vulnerability-contrast axis against the Assistant Axis, per grid layer. Random-cosine band ±0.033. The axes are near-orthogonal throughout.

cosine with $\hat{a}$	ℓ=20	ℓ=30	ℓ=40	ℓ=50	ℓ=60
$\hat{u}$ (RESP-MEAN PC1)	−0.066	−0.035	+0.000	+0.081	+0.118
$\hat{u}_{\text{LAST-USER}}$	+0.059	+0.097	+0.101	+0.111	+0.115
contrast axis	+0.149	+0.136	+0.143	+0.197	+0.226

Table 3: Tier-2 (independent IVs), persona level (n=150), ℓ=40. Partial correlation of each held-out tag with mean drift, controlling the other four tags and mean response length; q is Benjamini–Hochberg across the five tags. Joint-OLS β (all tags z-scored + length) in the last column (R^2=0.377). Vulnerability is null; the movers are expertise (+) and tech-literacy (−).

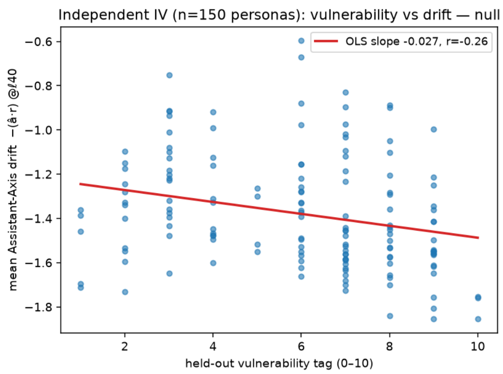
tag	**partial r**	p	q_{FDR}	**joint-OLS β**
expertise	+0.389	$< 10^{-3}$	$< 10^{-3}$	+0.123
tech_literacy	−0.452	$< 10^{-3}$	$< 10^{-3}$	−0.134
trust	−0.189	0.023	0.038	−0.048
emotional_load	−0.109	0.194	0.242	−0.037
vulnerability	−0.051	0.541	0.541	−0.022

Table 4: Tier-2 vulnerability→drift, rollout level (n=6,770): β_{vuln} (per z-unit, controlling length/arm/probe) across the layer grid and per arm, with bootstrap 95% CIs. Negative everywhere; the vuln×arm interaction is highly significant (β=+0.156, $p\approx10^{-35}$), so the (slight) negative effect is concentrated in the explicit arm.

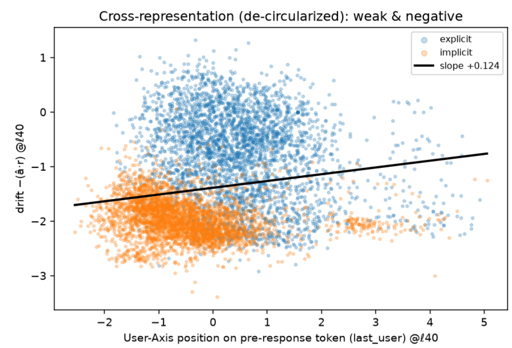
	ℓ=20	ℓ=30	ℓ=40	ℓ=50	ℓ=60
β_{vuln} (all)	−0.046	−0.066	−0.111	−0.190	−0.246
	explicit		implicit		
β_{vuln} @ℓ=40	−0.197 [−0.272, −0.126]		−0.025 [−0.037, −0.013]		

Table 5: Tier-3 (cross-representation): partial r of each user-position predictor with drift, controlling length/arm/probe (rollout, n=6,770) and length only (persona, n=150). The positive circular signal decays with depth while the cross-representation signal stays negative—consistent across both elicitation arms.

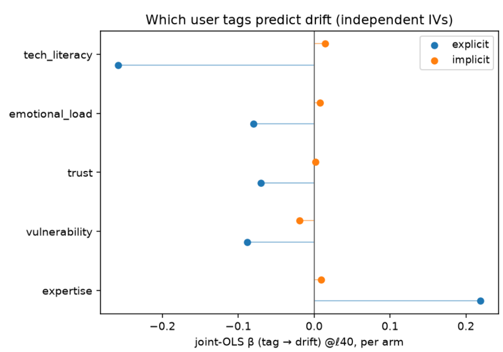
rollout partial r	ℓ=20	ℓ=30	ℓ=40	ℓ=50	ℓ=60
`userpos_resp` (circular)	+0.379	+0.324	+0.223	+0.042	−0.066
`userpos_lastB` (cross)	−0.147	−0.151	−0.163	−0.244	−0.258
@ℓ=40	resp	lastA	lastB		
rollout partial r	+0.223	+0.046	−0.163		
persona partial r (n=150)	+0.111	+0.096	−0.135		
explicit arm (rollout)	+0.159	—	−0.200		
implicit arm (rollout)	+0.163	—	−0.109		



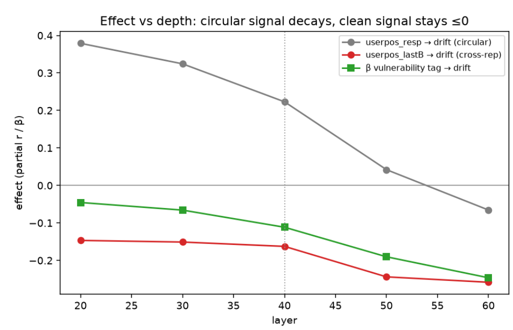
(a) Independent IV, persona level.



(b) Cross-representation, rollout level.



(c) Tag→drift coefficients per arm.



(d) Effect vs. layer.

Figure 2: Supporting figures for the link analysis. (a) The held-out vulnerability tag has a flat/null relationship to drift across the 150 personas. (b) The cross-representation predictor is weakly *negatively* related to drift. (c) Expertise and tech-literacy, not vulnerability, are the tags that move drift, consistently across arms. (d) The circular signal decays with depth while the clean signals stay ≤ 0.